Reconfigurable Dataflow Topology and the Hardware-Grounding Alignment Problem: A Structural Reformulation and Research Agenda for CGLA-class Architectures

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Abstract

Current AI accelerator design is optimized for the computational demands of large language models, but a structural question has received insufficient attention: does the topology of a hardware architecture constrain the class of sensory groundings it can faithfully implement? We propose a reformulation of this question using a topological framework introduced in a companion paper [**?**], which formalizes symbol grounding as a structural property of sensory space and introduces the Gromov–Hausdorff cognitive distance d_{cog} as a measure of grounding alignment between cognitive systems.

Within this framework, we argue that fixed-dataflow accelerators —including TPUs, GPUs, and memory-integrated SoCs such as the Apple M-series—impose a topological constraint on the class of sensory groundings they can implement, incurring a strictly positive d_{cog} lower bound for sensory spaces not embeddable in their fixed computational graph topology. We further propose that CGLA-class reconfigurable architectures are *candidates* for the architectural property required for universal sensory grounding, though this claim rests conditionally on an unproven conjecture (Section 4.2) whose verification or refutation is the central open obligation of this research program.

This paper does not present a completed theory. It presents a structural reformulation of the hardware-grounding alignment problem, identifies the key unresolved challenge for CGLA-class architectures, and proposes a research agenda connecting hardware topology to sensory topology. The necessary empirical and formal completion of this program—including verification of CGLA topological universality and empirical measurement of d_{cog} across hardware substrates—requires collaboration between cognitive theory and hardware architecture research. We close with five open problems at this intersection.

Contents

1 Introduction

The dominant AI accelerator paradigm is shaped by the computational demands of transformer-based large language models: massive matrix multiplication, high memory bandwidth, and fixed dataflow optimized for dense linear algebra. Google's Tensor Processing Unit (TPU) represents the extreme of this design philosophy—a systolic array architecture in which the dataflow between processing elements is fixed at design time [**?**]. This fixity yields exceptional performance per watt for the specific computational graphs of transformer inference, but it purchases efficiency at the cost of architectural flexibility.

A parallel line of inquiry has asked what *cognitive* requirements AI hardware should satisfy. If the goal of an AI system is not merely to process text efficiently, but to implement something resembling genuine understanding—grounded in sensorimotor experience and capable of concept formation across diverse domains—then the hardware substrate may matter in ways that current accelerator design does not address.

In a companion paper [**?**], we proposed a structural reformulation of symbol grounding: the topology of the sensory space in which grounding occurs may determine the geometry of the resulting cognitive representations, and two agents' grounding-induced topologies being homeomorphic is proposed as a sufficient condition for full intensional concept sharing. This reformulation of Harnad's symbol grounding problem [**?**] from a binary to a structural condition, if correct, has direct implications for hardware design.

The central proposal of the present paper is:

> *Fixed-dataflow architectures impose a topological restriction on implementable sensory spaces—this is the paper's primary formal claim (Proposition 1, conditional on Definition 5). We further propose that CGLA-class reconfigurable architectures are strong candidates for the architectural property required for universal sensory grounding, but this claim rests on Conjecture 1, which remains unproven. This paper's goal is to make these structural claims precise, identify what verification would require, and propose a research agenda for CGLA-class architecture as a candidate substrate for grounding-faithful AI systems.*

We develop this proposal in three steps. Section 2 reviews the framework from [**?**]. Section 3 argues that fixed dataflow implies topological restriction under Definition 5. Section 4 presents CGLA-class reconfigurability as a structural candidate and compares it qualitatively to FPGAs and fixed-dataflow alternatives. Section 5 discusses implications and introduces d_{cog} as a hardware selection criterion. Section 6 lists open problems that the CGLA research community is best positioned to resolve.

Remark 1 (On the status of this paper)**.** *This paper is a* foundational position paper*, not a completed theory. Its contribution is conceptual: it proposes a structural reformulation of the hardware-grounding alignment problem, identifies the topological restriction imposed by fixed-dataflow architectures, and specifies what CGLA-class architectures would need to demonstrate in order to satisfy the universality condition. The formal completion of the program—proof or disproof of Conjecture 1, empirical measurement of d_{cog}, and verification that CGLA configurations faithfully implement biological sensory topologies—requires collaboration between cognitive theory and hardware architecture research, and is beyond the scope of this paper alone.*

2 Background: Sensory Space Topology and Grounding

We briefly summarize the framework of [**?**], to which we refer the reader for full definitions and proofs.

Definition 1 (Sensory Space [**?**]). *The* sensory space *of a cognitive agent $\mathcal{A}$ equipped with transduction channels $\{T_1, \ldots, T_n\}$ is the product space*

$$\mathcal{S}(\mathcal{A}) = \prod_{i=1}^{n} A_i$$

equipped with the product topology, where each A_i carries the topology induced by the neural activation metric of channel T_i.

Definition 2 (Grounding Map [**?**]). *Let $\mathcal{K}(\mathcal{S}(\mathcal{A}))$ denote the hyperspace of nonempty closed subsets of $\mathcal{S}(\mathcal{A})$, equipped with the Vietoris topology [**?**]. A* grounding map *is a function*

$$G : \Sigma \to \mathcal{K}(\mathcal{S}(\mathcal{A}))$$

where Σ is a symbol set and $G(\sigma)$ is the closed set of sensory states associated with symbol σ.

Definition 3 (Grounding-Induced Topology [**?**]). *The* grounding-induced topology *$\tau_{\mathcal{A}}$ on Σ is the initial topology with respect to G: the coarsest topology making $\sigma \mapsto G(\sigma)$ continuous with respect to the Vietoris topology on $\mathcal{K}(\mathcal{S}(\mathcal{A}))$.*

The key result of [**?**] is that homeomorphism of grounding-induced topologies is *sufficient* for complete intensional concept sharing (Conjecture 1 of [**?**]), while topological inequivalence implies incommensurability of intensional content. The Gromov–Hausdorff cognitive distance

$$d_{\text{cog}}(\mathcal{A}, \mathcal{B}) = d_{GH}((\Sigma, \tau_{\mathcal{A}}),\ (\Sigma, \tau_{\mathcal{B}}))$$

measures the degree of grounding alignment between two agents, with $d_{\text{cog}} = 0$ corresponding to the homeomorphic (complete sharing) case [**?**].

3 Fixed Dataflow as Topological Restriction

3.1 Computational Graphs and Their Topology

A neural network computation can be represented as a directed acyclic graph $\mathcal{G} = (V, E)$ where vertices V are processing elements and directed edges E represent dataflow. The *computational graph topology* is the topology induced on V by the adjacency structure of $\mathcal{G}$.

Definition 4 (Fixed Dataflow Architecture). *An architecture is* fixed-dataflow *if the computational graph $\mathcal{G}$ is determined at design time and cannot be reconfigured during operation. The topology of $\mathcal{G}$ is therefore a constant of the hardware.*

TPUs are fixed-dataflow in this sense: the systolic array defines a rigid grid topology in which data flows in a fixed pattern optimized for matrix multiplication. Alternative operations—including those arising from non-matrix sensory modalities—must be mapped onto this fixed topology, potentially at significant computational cost or loss of structural fidelity.

3.2 The Topological Restriction Proposition

The connection between sensory space topology and computational graph topology is the following. Any implementation of a grounding map G on hardware requires a mapping from the topological structure of $\mathcal{S}(\mathcal{A})$ to the topological structure of the computational graph $\mathcal{G}$. We formalize this as follows.

Definition 5 (Topological Implementation Map)**.** *A* topological implementation map *from sensory space* $(\mathcal{S}(\mathcal{A}), \tau_{\mathcal{A}})$ *to a hardware architecture* $\mathcal{H}$ *with computational graph* $(\mathcal{G}, \tau_{\mathcal{G}})$ *is a continuous injection*

$$\phi : (\Sigma, \tau_{\mathcal{A}}) \hookrightarrow (\mathcal{G}, \tau_{\mathcal{G}}).$$

The implementation is faithful *if* ϕ *is a homeomorphism onto its image, i.e., both* ϕ *and* ϕ^{-1} *preserve the topological structure of the grounding-induced symbol space.*

Remark 2 (Conceptual justification of Definition 5)**.** *Definition 5 maps between two spaces of fundamentally different character:* $(\Sigma, \tau_{\mathcal{A}})$ *is a semantic space—the symbol set equipped with the topology induced by sensorimotor grounding —while* $(\mathcal{G}, \tau_{\mathcal{G}})$ *is a physical space—the processing element graph of a hardware chip. This cross-domain mapping requires justification.*

The conceptual bridge is the following: any physical implementation of a grounding map $G : \Sigma \to \mathcal{K}(\mathcal{S}(\mathcal{A}))$ *must assign each symbol* $\sigma \in \Sigma$ *to a set of computational states that represent* $G(\sigma)$*, and must implement the relational structure among symbols as relational structure among computational processes. Symbols that are topologically close in* $\tau_{\mathcal{A}}$ *—meaning their grounding sets are similar in sensory space—should correspond to computationally related processes in* $\mathcal{G}$*. If the computational graph lacks the relational structure to place such processes in proximity, the topological relationships among grounded symbols are distorted. Definition 5 formalizes this requirement as continuity:* ϕ *preserves the topological relationships of the semantic space in the computational graph.*

The faithfulness condition (homeomorphism onto image) further requires that no relational structure is lost in either direction: neither is $\tau_{\mathcal{A}}$*-proximity collapsed into* $\tau_{\mathcal{G}}$*- distance (loss of discrimination), nor is* $\tau_{\mathcal{G}}$*-proximity imposed on* $\tau_{\mathcal{A}}$*-distant symbols (spurious conflation).*

We acknowledge that this definition conflates semantic topology and physical topology in a way that requires empirical grounding—namely, the specification of what metric on $\mathcal{G}$ *is relevant for cognitive purposes. This is OP1; the formal adequacy of Definition 5 is therefore conditional on its resolution.*

The following restriction holds as a direct consequence of Definition 5; its force depends on the adequacy of that definition, which we identify as the most fundamental open problem of this framework (OP1).

Proposition 1 (Topological Restriction of Fixed Dataflow (under Definition 5))**.** *Let $\mathcal{H}$ be a fixed-dataflow architecture with computational graph topology $\tau_{\mathcal{G}}$. Then $\mathcal{H}$ can implement only grounding maps G whose induced topology $\tau_{\mathcal{A}}$ is homeomorphic to a subspace of $(\mathcal{G}, \tau_{\mathcal{G}})$. Sensory spaces whose topology is not homeomorphic to any subspace of $(\mathcal{G}, \tau_{\mathcal{G}})$ cannot be faithfully grounded on $\mathcal{H}$.*

Remark 3. *We emphasize the distinction between* computational universality *and* topological faithfulness*. A fixed-dataflow architecture is Turing-complete in the sense that it can compute any computable function; the restriction here is not on computability but on structural preservation. When a grounding map G whose sensory topology $\tau_{\mathcal{A}}$ is not embeddable in $\tau_{\mathcal{G}}$ is executed on $\mathcal{H}$, the computation proceeds correctly but the topological relationships among grounded symbols are distorted. Zero-padding and format-conversion workarounds preserve the numerical output of individual computations while destroying the relational structure of the grounding space. This distortion is therefore structural, not merely a performance overhead: it implies $d_{\text{cog}} > 0$ as a* lower bound *on the achievable cognitive distance, not a recoverable inefficiency.*

3.3 Concrete Case: TPU and Chemical Sensory Space

Consider the sensory space of *Caenorhabditis elegans*: a low-dimensional chemical topology dominated by volatile compound gradients [**?**]. The natural computational graph for implementing chemotaxis-based grounding is a sparse, gradient-following structure with few processing elements and sequential update dynamics [**?**].

The TPU's systolic array, by contrast, is optimized for dense $n \times n$ matrix products. Mapping a sparse gradient-following computation onto a dense matrix architecture requires either zero-padding (wasting compute) or restructuring the computation to fit the matrix template (distorting the grounding topology). Neither preserves the topological structure of the original sensory space.

This is a concrete instance of topological restriction: the TPU's fixed topology is well-matched to transformer attention but poorly matched to chemical gradient computation. Any agent implemented on a TPU that attempts to ground concepts in a chemical sensory space will suffer elevated d_{cog} relative to a native chemical-topology implementation.

3.4 Contrast Case: GPU and Kernel-Level Pseudo-Flexibility

Graphics Processing Units (GPUs) appear more flexible than TPUs: CUDA kernels can be selected at runtime to implement diverse computational patterns, and the programmer can target different memory access patterns and parallelization strategies depending on the workload. This might suggest that GPUs occupy an intermediate position in topological flexibility between TPUs and CGLA-class architectures.

We argue this appearance is misleading. GPU flexibility resides at the *software layer*—kernel selection determines which computation executes on the hardware, but does not change the hardware topology itself. The underlying physical architecture is a fixed grid of Streaming Multiprocessors (SMs) connected through a hierarchical memory system (L1

cache, L2 cache, HBM). This hardware topology is determined at design time and is not reconfigurable during operation.

Under Definition 3, GPUs are therefore fixed-dataflow in the relevant sense: the computational graph topology $\tau_{\mathcal{G}}$ of the hardware remains constant across kernel invocations. A sparse gradient-following computation (such as the *C. elegans* chemotaxis example of Section 3.2) mapped onto a GPU requires the same structural translation as on a TPU: the sparse graph must be padded or restructured to fit the dense SM grid topology. The resulting d_{cog} lower bound is therefore qualitatively identical to the TPU case, differing in magnitude depending on the degree of sparsity mismatch rather than in kind.

This distinction—between software-layer flexibility and hardware-topology reconfigurability—is the critical boundary separating fixed-dataflow and reconfigurable-dataflow architectures for the purposes of grounding fidelity. The CGLA's reconfigurable interconnect changes the actual computational graph topology $\tau_{\mathcal{G}}$; GPU kernel selection does not.

We note that Field-Programmable Gate Arrays (FPGAs) also offer hardware-level reconfigurability and thus do not fall under the fixed-dataflow restriction of Proposition 1. The comparative analysis of d_{cog}-minimization between CGLA-class and FPGA architectures is deferred to OP5; the present argument is that reconfigurability is a necessary architectural property for universal grounding, not that CGLA is its unique instantiation.

3.5 Extended Case: Apple M-series and Memory-Integrated Fixed Dataflow

Apple's M-series System-on-Chip (SoC) architecture represents a distinct and instructive case that differs from both TPU and GPU in its integration strategy. The M5 chip [**?**] integrates a 16-core Neural Engine, a 10-core GPU with a dedicated Neural Accelerator in each core, and a CPU, all sharing a unified memory pool via a high-bandwidth interconnect (153 GB/s). This unified memory architecture eliminates data transfer overhead between processing units—a genuine engineering advantage over discrete chip configurations.

However, the topological analysis of the present framework reveals that memory integration and dataflow reconfigurability are orthogonal properties. The M-series Neural Engine is a fixed-dataflow accelerator: its computational graph topology is determined at design time and optimized for matrix-based neural network operations. The Neural Accelerators within each GPU core are similarly fixed-topology units. The unified memory architecture addresses the *bandwidth* bottleneck between processing units, but does not address the *topological mismatch* between the hardware's fixed computational graph structure and the target sensory space topology.

Concretely: an M5 implementation of *C. elegans* chemotaxis faces the same topological restriction as a TPU implementation. The sparse, gradient-following computational graph required by the chemotaxis loop must be mapped onto fixed matrix-operation units, incurring the same structural distortion that elevates d_{cog}. The unified memory reduces the cost of transferring intermediate representations between CPU, GPU, and Neural Engine, but the representations themselves are computed in fixed-topology units that impose the same homeomorphism constraint as Proposition 1.

Remark 4 (Memory Integration vs. Topological Flexibility)**.** *This distinction is architec-*

turally fundamental. Memory-integrated SoC designs (Apple M-series, Qualcomm Snapdragon) optimize the communication *topology between fixed-dataflow processing units. CGLA-class reconfigurable architectures optimize the* computational *topology of the processing units themselves. For the purposes of d_{cog}-minimization, only the latter is relevant: the cognitive distance between two grounding implementations is determined by the computational graph topology available to implement the grounding map G, not by the memory bandwidth available to transfer activations between stages. A system with high memory bandwidth and fixed computational topology will achieve lower d_{cog} than a low-bandwidth system with the same fixed topology—but a system with reconfigurable computational topology can achieve $d_{\text{cog}} \to 0$ for sensory spaces within its configuration space, regardless of memory bandwidth.*

4 CGLA-class Reconfigurability as Topological Flexibility

4.1 CGLA Architecture

The Coarse-Grained Linear Array (CGLA) is a reconfigurable computing architecture in which the dataflow between processing elements can be modified at runtime [**?**, **?**]. Unlike the systolic array's fixed grid, CGLA allows the computational graph topology $\tau_{\mathcal{G}}$ to be reconfigured to match the structure of the workload.

Recent work has demonstrated CGLA implementations of LLM inference [**?**], attention acceleration [**?**], and energy-efficient whisper ASR [**?**], showing that the architecture can handle diverse computational structures within a single hardware substrate. The key property for our purposes is not performance per se, but *topological flexibility*: the ability to present different computational graph topologies to different workloads.

We note that other reconfigurable and neuromorphic architectures— including Intel's Loihi 2 [**?**], SpiNNaker [**?**], and BrainScaleS [**?**]—also offer topological flexibility via configurable spike-routing networks. The present argument applies to the class of reconfigurable-dataflow architectures broadly; CGLA is the primary instantiation we consider because its hardware-efficiency properties under diverse computational workloads have been most recently documented [**?**, **?**]. A systematic comparative analysis of d_{cog}-minimization across these architectures is deferred to OP5.

4.2 Topological Flexibility and Universal Grounding

Definition 6 (Topologically Universal Architecture)**.** *An architecture $\mathcal{H}$ is* topologically universal *with respect to a class $\mathcal{C}$ of sensory spaces if for every $\mathcal{S} \in \mathcal{C}$, there exists a configuration of $\mathcal{H}$ whose computational graph topology is homeomorphic to the grounding-induced topology of $\mathcal{S}$.*

Conjecture 1 (CGLA Topological Universality)**.** *CGLA-class architectures are topologically universal with respect to the class of sensory spaces arising from biological transduction channels (chemical, visual, mechanoreceptive, proprioceptive, and their cross-modal products).*

Remark 5 (On the strength of the homeomorphism condition)**.** *The universality conjecture requires exact homeomorphism between CGLA computational graph topologies and biological sensory topologies. This is a strong condition that may not hold in practice, for two reasons.*

*First, biological sensory spaces are continuous, high-dimensional, and fiber-bundle-structured in many organisms (Section 3.2 of [**?**]), while CGLA computational graphs are finite and discrete. A faithful homeomorphism in the strict sense therefore requires discretization, which necessarily involves approximation.*

Second, even if CGLA's configuration space is dense in the metric space of sensory topologies (OP5), there may be sensory topologies that require configurations exceeding CGLA's physical connectivity constraints.

We therefore propose a weakened version of the universality condition as an alternative target for verification: rather than exact homeomorphism, require that CGLA can achieve d_{cog} below a threshold $\epsilon > 0$ for all sensory spaces in the biological class. This ϵ-universality condition is more empirically tractable and biologically meaningful, since perfect grounding fidelity is not required for functional cognitive equivalence. We flag this weakening as part of OP2.

The conjecture is supported by the following structural argument. The CGLA's reconfigurable interconnect allows the processing element graph to be set to any sparse directed graph within the physical connectivity constraints of the chip. The sensory spaces of biological organisms, while high-dimensional, are structured by the specific connectivity patterns of neural circuits—which are themselves sparse and locally connected [**?**, **?**]. A reconfigurable architecture that can present sparse, locally-connected computational graphs is therefore structurally aligned with biological sensory topologies in a way that dense matrix architectures are not.

Remark 6 (Conditional Necessity)**.** *The paper's central claim—reconfigurable dataflow topology as a* necessary *condition—rests conditionally on this conjecture. What follows from Proposition 1 alone is that fixed-dataflow architectures impose a strictly positive lower bound on d_{cog} for sensory spaces not embeddable in $\tau_{\mathcal{G}}$. The necessity claim then holds as follows: if Conjecture 1 holds, then CGLA-class reconfigurability achieves a lower bound of zero, making it the unique class of physical architectures capable of faithful universal grounding. Proof of Conjecture 1 (OP2) is therefore the most pressing remaining obligation of this research program.*

4.3 CGLA vs. FPGA: A Qualitative Comparison

As noted in Section 3.3, FPGAs also offer hardware-level reconfigurability and are therefore not subject to the fixed-dataflow restriction of Proposition 1. This raises a legitimate question: is CGLA's advantage over fixed-dataflow architectures shared equally by FPGAs, and if so, what is CGLA's distinctive contribution?

We offer a qualitative comparison on three dimensions relevant to d_{cog}-minimization.

Reconfiguration granularity. FPGAs achieve reconfigurability via lookup tables (LUTs) and programmable interconnects at the gate level, enabling arbitrary logic functions. CGLA achieves reconfigurability at the level of coarse-grained functional units (arithmetic and logical operations) connected by a reconfigurable routing network [**?**]. For the purpose of im-

plementing sensory grounding topologies—which are structured by sparse, directed graphs of moderate complexity rather than arbitrary Boolean circuits—CGLA's coarse-grained reconfigurability is well-matched to the target structure. FPGA's fine-grained LUT-level programming introduces significant overhead (both area and latency) for the kind of structured sparse computations that biological sensory topologies require. In this sense, CGLA is a "right-sized" reconfigurable architecture for grounding-topology implementation, whereas FPGA is over-specified for this task and incurs avoidable energy costs.

Energy efficiency. CGLA's empirically demonstrated energy efficiency [**?**, **?**] reflects this granularity match: reconfiguring at the functional-unit level consumes less power than reconfiguring at the gate level for computations that do not require gate-level flexibility. For grounding implementations intended to run continuously in embodied systems—where energy efficiency is a hard constraint—this distinction is practically significant.

Scope of the present argument. The claim of the present paper is that *reconfigurable dataflow topology is necessary* for universal grounding, not that CGLA is the unique architecture satisfying this condition. FPGAs satisfy the topological reconfigurability condition and should in principle achieve $d_{\text{cog}} \to 0$ for sensory spaces within their configuration space. The comparative analysis of d_{cog}-minimization efficiency between CGLA and FPGA implementations is deferred to OP5; the present argument establishes the necessary property, and CGLA is the primary instantiation we consider because its efficiency under diverse workloads has been most recently and rigorously documented [**?**, **?**].

5 Implications

5.1 Hardware Selection as Cognitive Design

The framework developed here implies that hardware selection is not merely a performance engineering decision but a cognitive design decision. An AI system intended to implement human-like grounding requires hardware whose computational graph topology can approximate the topology of human sensory space. A system grounded in a chemical sensory space (like a robotic nose) requires hardware whose topology approximates a sparse gradient-following graph.

The cognitive distance d_{cog} provides a principled criterion for hardware selection: given a target sensory topology $\tau_{\mathcal{A}}$, select the hardware architecture $\mathcal{H}$ that minimizes d_{cog} between $\tau_{\mathcal{A}}$ and the best achievable computational graph topology of $\mathcal{H}$.

For transformer-based LLMs with attention-pattern topology, this criterion favors TPUs. For systems intended to implement diverse sensory groundings, this criterion favors CGLA-class architectures.

5.2 Neuro-Symbolic Alignment

Neuro-symbolic AI systems combine a neural perception layer with a symbolic reasoning layer [**?**]. The topological alignment condition of [**?**]—that the symbolic layer's induced topology should be consistent with the neural layer's grounding topology—extends naturally to hardware: the hardware topology of the neural layer should be consistent with the sensory

topology, and the hardware topology of the symbolic layer should be consistent with the relational structure of the symbol set.

CGLA-class architectures, capable of presenting different topologies to different layers of a neuro-symbolic system, are therefore structurally suited to implementing topological alignment at the hardware level.

5.3 Energy Efficiency as a Consequence of Topological Matching

A secondary implication concerns energy efficiency. When hardware topology is mismatched to computational topology, energy is expended on structural translation: zero-padding, reshaping, and format conversion operations that carry no semantic content. Topological matching eliminates this overhead.

This connects to CGLA's empirically demonstrated energy efficiency advantages [**?**]: the efficiency gain is not merely an engineering achievement but a structural consequence of reduced topological mismatch between the computation and the hardware.

6 Pilot Evidence: RSA as d_{cog} Proxy in LLMs

Remark 7 (Which component of d_{cog} does RSA measure?)**.** *Under the persistence signature formulation of [**?**, Def. **??**], d_{cog} is a tuple whose first two components (d_{GH} and $d_{\mathrm{bn}}^{(0)}$) are computable from an RSA-derived metric on Σ. The second-order Spearman ρ_{cog} used below is a monotone proxy for $d_{\mathrm{bn}}^{(0)}$ — it tracks merge-structure alignment, not loop structure. Higher-order components $d_{\mathrm{bn}}^{(k)}$ ($k \geq 1$) are not resolved at the twenty-nine–concept inventory used here (the detection threshold is $N \gtrsim 24$; see [**?**, App. **??**]). The pilot below therefore substantiates the H_0-level proxy claim; the H_1-level remains open.*

OP4 proposes RSA as a tractable proxy for d_{cog} in toy hardware experiments. We report a pilot study applying this proxy to language model training checkpoints, providing a software-level proof of concept for the RSA-based d_{cog} measurement protocol.

Setup. We computed RSA dissimilarity matrices over $n = 29$ probe concepts at hidden layers $\ell \in \{2, 4, 6\}$ of Pythia-70m [**?**] at three training checkpoints: step 0 (random initialization), step 1000 (early training), step 143000 (fully trained reference). Three prompt templates were averaged to reduce surface-form bias. Structural alignment to the reference was measured as the second-order Spearman correlation ρ_{cog}, which OP4 proposes as the operative proxy for d_{cog}.

Results. ρ_{cog} increases monotonically with training at all layers (L6: step-0 $= 0.017$; step-1000 $= 0.535$), confirming that RSA captures a meaningful structural signal that evolves predictably with training. Cross-model replication with GPT-2 [**?**] shows $\rho_{\mathrm{cog}}(\text{GPT-2}, \text{Pythia-trained}) \approx 0.42$ versus $\rho_{\mathrm{cog}}(\text{GPT-2}, \text{Pythia-step-0}) \approx 0.10$, demonstrating that ρ_{cog} discriminates structurally distinct representations.

Relevance to hardware experiments. This result supports the validity of the RSA proxy for the hardware d_{cog} measurement proposed in OP3–OP4. Specifically, it demonstrates that: (i) ρ_{cog} is sensitive to structural differences in representational geometry at fixed architecture; (ii) the proxy is stable across prompt formulations when averaged; (iii) cross-architecture comparison (GPT-2 vs. Pythia) is feasible with the same protocol.

The analogous hardware experiment—comparing RSA dissimilarity matrices of grounded representations on TPU, GPU, and CGLA substrates—would use the same ρ_{cog} protocol, substituting hardware-level activation traces for model checkpoint activations. The prediction of Section 5 is that CGLA implementations yield higher ρ_{cog} to the target topology than fixed-dataflow substrates at matched computational budget. All code is available at `https://github.com/minamominamoto/grounding-without-reality`.

7 Open Problems

OP1. Formal definition of topological implementation map. Define precisely the mapping from sensory space topology to computational graph topology. This requires specifying which properties of $\tau_{\mathcal{A}}$ are preserved under discretization onto a finite processing element graph, and which are necessarily lost.

OP2. Proof or disproof of CGLA Topological Universality. The conjecture in Section 4.2 requires a formal proof that CGLA's reconfigurable interconnect can present computational graph topologies homeomorphic to the grounding-induced topologies of biological sensory spaces. A disproof would require exhibiting a biologically motivated sensory topology that cannot be approximated by any CGLA configuration within given physical constraints.

OP3. Empirical measurement of d_{cog} on hardware. Given a fixed sensory grounding task (e.g., chemical gradient following in a simulated *C. elegans* environment), measure d_{cog} between implementations on TPU, GPU, and CGLA-class hardware. The prediction of the present paper is that CGLA achieves the smallest d_{cog} at comparable energy expenditure.

OP4. Toy model of d_{cog}. As a minimal concrete instantiation prior to full hardware experiments, consider a two-dimensional chemical sensory space $\mathcal{S}_{\text{chem}}$ and a three-dimensional visual sensory space $\mathcal{S}_{\text{vis}}$, each with synthetically generated grounding maps. Representational Similarity Analysis (RSA) [**?**] provides an operational proxy for d_{cog}: compute the dissimilarity matrix of grounded representations on each hardware substrate and compare via Spearman correlation to the target topology's dissimilarity matrix.

On the validity of RSA as a proxy for d_{cog}. RSA operates on pairwise dissimilarity matrices and is therefore sensitive to the ordinal structure of representational similarity rather than to absolute distances. Its Spearman correlation with a target dissimilarity matrix approximates d_{cog} under the assumption that the sensory space is *locally isometric* in the regions of interest—that is, that the ratio of pairwise distances is preserved under the grounding map G in a neighbourhood around each grounded

concept. When this assumption holds, RSA similarity correlates inversely with d_{cog}: a higher Spearman correlation indicates smaller topological distortion. When the sensory space is highly non-isometric (e.g., fiber bundle structures as in *Drosophila*), RSA may underestimate d_{cog} because it does not capture global topological invariants such as connectivity or homotopy type. In such cases, persistent homology methods would provide a more faithful proxy. For the purposes of the toy model (chemical and visual sensory spaces with product topology), the local isometry assumption is plausible and RSA is a tractable first approximation.

The prediction is that, for non-matrix sensory spaces at matched computational budgets: (i) CGLA implementations yield higher RSA similarity to the target topology than both TPU and GPU implementations; (ii) TPU and GPU implementations yield qualitatively similar RSA similarity to each other, differing in magnitude but not in kind, since both are fixed at the hardware topology layer (Section 3.3); and (iii) the GPU–CGLA gap exceeds the TPU–GPU gap, because GPU software-layer flexibility does not reduce the structural translation cost that drives d_{cog}.

OP5. Extension to cross-modal sensory spaces. The *Drosophila* central complex exhibits cross-modal convergence that cannot be captured by a product topology [**?**]. Extending the framework to fiber bundle sensory spaces and determining whether CGLA's interconnect can represent fiber bundle computational graphs is an open problem.

OP6. Hardware topology metric. Define a metric on computational graph topologies analogous to d_{cog}, and establish whether the CGLA's configuration space is dense in this metric space (i.e., whether it can approximate any target computational graph topology to arbitrary precision within physical constraints).

8 Conclusion

We have proposed a structural reformulation of the hardware-grounding alignment problem. Within the topological framework of [**?**], fixed-dataflow architectures incur a strictly positive d_{cog} lower bound for sensory spaces not embeddable in their fixed computational graph topology—this is the paper's primary formal claim, conditional on Definition 5. We have further proposed that CGLA-class reconfigurable architectures are strong candidates for the architectural property required for universal sensory grounding, though this candidacy rests on Conjecture 1, which remains unproven.

This paper does not claim to have solved the hardware-grounding alignment problem. It claims to have made the problem formally precise. The research agenda that follows from this reformulation has three components.

For cognitive theory: Provide a formal proof or disproof of Conjecture 1 (OP2). Determine whether exact homeomorphism is the correct universality condition, or whether the ϵ-universality weakening is more appropriate (Section 4.2). Formalize Definition 5 (OP1) so that the topological implementation map has a rigorous operational meaning.

For CGLA architecture research: Characterize the configuration space of CGLA's

reconfigurable interconnect formally, and determine whether it is dense in the metric space of biological sensory topologies (OP5). Provide at least one concrete instantiation of Definition 5—a specific CGLA configuration that implements the *C. elegans* chemosensory topology—as a proof of concept. Measure d_{cog} between CGLA and fixed-dataflow implementations of the same sensory grounding task using RSA as a proxy (OP3, OP4).

For the intersection: As AI systems are increasingly deployed in embodied, multimodal, and neuromorphic contexts, the alignment between hardware topology and sensory topology will become a principled design criterion. The cognitive distance d_{cog} provides the formal foundation for this criterion. Realizing it as an engineering tool requires collaboration between the communities that this paper attempts to bridge.

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